ADVANCED SEGMENTED-VECTOR PULSE FREQUENCY MODULATION FOR THREE-LEVEL CONVERTER BASED WIRELESS POWER TRANSFER SYSTEM

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Abstract

Three-level converters controlled by an advanced segmented-vector pulse frequency modulation (A-SVPFM) strategy are promising for high-power wireless power transfer (WPT). This paper proposes an A-SVPFM method and a dedicated digital modulator for a half-bridge flying-capacitor three-level converter. The modulator maps a continuous normalized voltage ratio into adjacent whole-wave pulses and preserves a zero-switching angle at the resonant frequency across the control range, enabling zero-voltage switching (ZVS) with wide-range

A switching-state selection scheme that uses only voltage sampling is further introduced to regulate the flying-capacitor voltage, avoiding current sensors while maintaining small ripple around the reference.

On the analysis side, the Fourier series of the system output-voltage is formulated and the interharmonics structure induced by the finite modulation period is clarified; discrete Fourier transform (DFT) is then employed to compare A-SVPFM with pulse frequency modulation (PFM), showing a markedly lower transmitter-current total harmonic distortion (THD) under the band-pass input impedance of the WPT link.

Moreover, a closed-form ZVS condition is derived that links the required transmitter-capacitance offset, dead time, and load, providing practical design guidance.

The effectiveness of the proposed scheme is validated by software simulations and hardware experiments on a three-level WPT prototype.

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Chapter 1

Introduction

Wireless power transfer (WPT) technology, started from the work of Nikola Tesla over a century ago, it has a wide range of application scenarios, such as micro-integrated systems to electronic medical devices [1], [2]. One of the important application scenarios of wireless charging technology is charging electric vehicles [4], the WPT technology provides innovative performance in wide range of aspects like: convenience, electrical isolation, and user experience [4].

Multilevel inverters have emerged as a promising alternative to conventional two-level topologies in wireless power transfer (WPT) systems, primarily due to their superior voltage handling capability and adaptability to multichannel configurations. Multilevel inverters provide a feasible solution by enabling higher voltage operation while simultaneously supporting decoupled multichannel WPT architectures, which eliminate bulky transformers and simplify system design [5], [6]. Moreover, multilevel inverters inherently offer reduced low-order harmonic distortion, lower device voltage stress, and enhanced reliability [7].

A variety of modulation strategies have been explored for two-level inverter-based WPT systems, such as pulse width modulation (PWM), pulse density modulation (PDM), pulse frequency modulation (PFM), pulse magnitude modulation (PMM) [8], and hybrid approaches. Among them, phase-shift control [9], [10], which can be regarded as a PWM variant with an adjustable phase angle, is commonly adopted for regulating converter operation in WPT applications. However, this method is often associated with hard-switching conditions unless additional soft-switching techniques are incorporated. ON–OFF modulation, on the other hand, relies on low-frequency duty-cycle regulation of the converters [11], but its performance is limited by low average efficiency and significant output ripple. Hybrid modulation attempts to merge the concepts of PFM and ON–OFF control; nevertheless, it still suffers from pronounced ripple issues [12]. In contrast, both PDM and PFM are regarded as promising candidates for WPT systems, as they effectively address the drawbacks.

Pulse density modulation (PDM) was initially introduced for dual-side half-bridge converter-based WPT systems in [13], with the objectives of achieving zero-voltage switching (ZVS) and enabling maximum power tracking. In this scheme, the output power is adjusted by varying the density of pulses. Nonetheless, the soft-switching performance of PDM-based WPT systems is strongly affected by coupling conditions and load variations [11]. To enhance ZVS capability across a wider range of operating scenarios, a PDM full-bridge converter incorporating an additional ZVS branch between switching nodes was presented in [14]. While this configuration ensures ZVS, it introduces several drawbacks, including structural modifications to the inverter, the presence of low-frequency subharmonics, a restricted modulation range, and inherent modulation delays. To overcome these limitations, [15] proposed an advanced PDM scheme characterized by low subharmonic distortion, wide-range modulation capability, and fast dynamic response. Moreover, [16] and [17] proposed the shifting to half-bridge mode from full-bridge mode of advanced PDM, since the half-bridge mode provides only half of the output voltage compared to the full-bridge mode, the resulting output ripple is reduced by a factor of two. However, PDM-based WPT systems still require the incorporation of auxiliary circuits to alter the current frequency in order to reliably sustain soft switching.

Given these observations, there is an urgent need for further exploration of soft-switching modulation techniques tailored to multilevel inverter-based WPT systems, to fully exploit their advantages while mitigating switching losses.

To avoid the need for additional auxiliary circuitry, pulse frequency modulation (PFM) schemes for WPT systems were introduced in [18],[19],[20],[21]. In these approaches, the output power is controlled by exploiting the high-order harmonics of the square wave at designated frequencies. The concept of half-wave PFM was first reported in [18], [19], while a half-wave autonomous PFM was later developed for self-excited oscillating WPT systems, enabling power regulation with reliable ZVS operation [22]. Most of the works have concentrated on series-compensated topologies, studies in [23] further apply PFM to wireless charging with high-order compensation topologies.

A key drawback of such methods is that the number of hybrid-frequency pulse sequences must be explicitly determined for each modulation index. [24] proposed the Σ-Δ modulator of advanced PFM, where the frequency is changed directly by holding different sampling periods of the control system.

More recently, step density modulation was introduced in [25] from a different perspective. This technique adjusts the output power by varying the density of steps, while maintaining the same waveform characteristics as PFM. And [26] proposed a Σ-Δ modulator for SVPFM-WPT system, which can realize wide range ZVS without auxiliary circuit and inverter modification.

The researches above primarily address modulation strategies for inverters in WPT systems. In contrast, soft-switching modulation techniques for multilevel inverters remain insufficiently investigated in this context. A critical challenge associated with multilevel inverters is the issue of capacitor voltage balancing [27],[28],[29]. Therefore, considering the distinctive characteristics of WPT systems, it is essential to develop a straightforward yet effective approach to ensure capacitor voltage balance in multilevel inverter operation.

Compared to SVPFM, the proposed A-SVPFM allows continuously adjustable resolution λ, which enables intentional repositioning and compression of interharmonics near the resonance band, introducing degrees of freedom not present in SVPFM, directly reducing THD and current/voltage ripples. The same mechanism yields a shorter and explicitly controllable modulation period, leading to a cleaner spectrum and faster transient settling. Moreover, while both schemes are designed for soft-switching, A-SVPFM achieves the efficiency–harmonic co-optimization more effectively, maintaining high conversion efficiency across a wide operating range while simultaneously suppressing THD and dc-link ripple.

Driven by the objective of achieving soft-switching modulation in multilevel converter-based WPT systems, this work introduces an advanced segmented-vector pulse frequency modulation (A-SVPFM) approach. The principal contributions of this study can be summarized as follows:

1) An A-SVPFM technique is presented to achieve wide-range zero-voltage switching (ZVS) in half-bridge flying-capacitor based three-level converter for WPT applications.
2) A switching-state-based capacitor voltage balancing strategy is introduced for the flying-capacitor three-level half-bridge converter, enabling the flying capacitor voltage to be maintained at its reference value with minimal ripple.
3) A closed-loop condition relating the capacitive bias at the transmitter to the minimum phase/dead-time to the load is derived, which provides quantifiable design basis for ZVS coverage prediction and margin allocation.

For the remainder of this dissertation: chapter 2 introduces the fundamental principles of the proposed A-SVPFM for WPT systems; chapter 3 describes the capacitor voltage balancing technique for this 3-level converter that does not require current sensors; chapter 4 provides experimental validation of the A-SVPFM method. Finally; chapter 5 summarizes the conclusions of this study.

Chapter 2

Advanced SVPFM WPT system

2.1 System topology

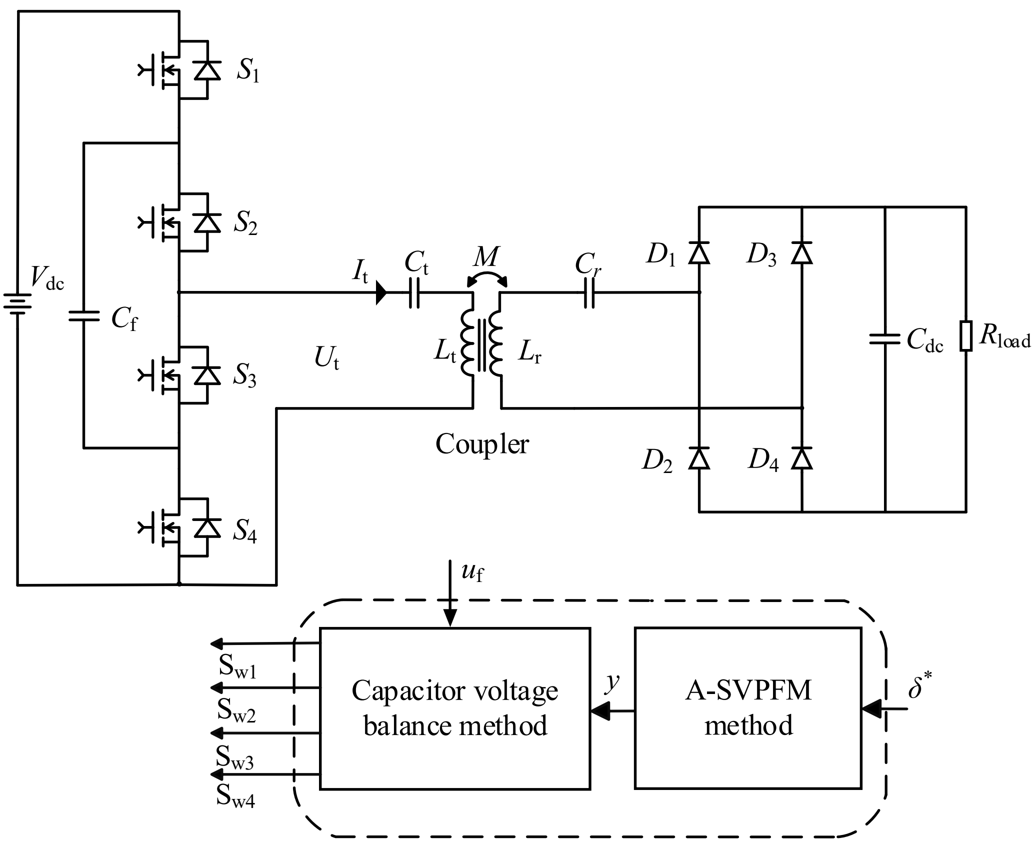

Figure 2.1: The A-SVPFM flying-capacitor three-level converter-based WPT system

Fig 2.1 shows the figure of advanced-SVPFM WPT system, the wireless power transfer system comprises a controller, a driving circuit, a sampling circuit, a DC voltage source (steady-state voltage: V_{dc}), a flying capacitor three-level converter, a transmitting compensation capacitor (C_t), a transmitting coil (), a receiving coil (L_r), a receiving compensation capacitor (C_r), and a diode rectifier (D_1~D_4), the DC voltage source is connected to the flying capacitor three-level converter. The flying-capacitor three-level converter consists of switching devices S_1, S_2, S_3, $S_4$and a flying capacitor C_f. These switches form upper and lower bridge arms, with the flying capacitor C_f connected in series between the upper and lower bridge arms.

The midpoint of the bridge arms are connected to the input of the transmitting compensation network (C_t). The transmitting compensation network (C_t), transmitting coil (L_t), receiving coil (L_r), receiving compensation network (C_r), diode rectifier (D_1~D_4), and DC-side output capacitor (C_{dc}) are sequentially connected to form the power transfer path. The power is transferred from the transmitter side to the receiver side via the mutual inductance M.

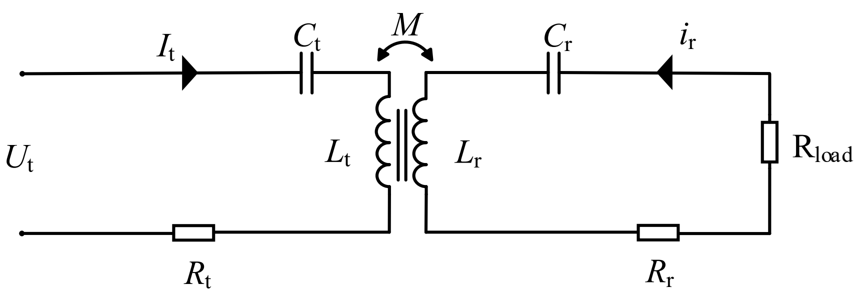

Fig.2.2 A simple equivalent circuit of WPT system

The three-level half-bridge converter operates at the resonant frequency of the LC resonator. The relationship between the transmitting and receiving LC resonators is expressed as follows:

$$f_r = \frac{1}{2\pi\sqrt{L_t C_t}} = \frac{1}{2\pi\sqrt{L_r C_r}} \tag{2.1}$$

where f_r is the resonant frequency, L_t and L_r are the self-inductances of the transmitting and receiving coupler, C_t and C_r are compensated capacitors on the transmitter and receiver sides, respectively.

The series-series compensated WPT system circuit equations can be given as:

$$u_t=\left(j\omega\, L_t+\frac{1}{j\omega C_t}+R_t\right)i_t+j\omega M i_r \tag{2.2}$$

$$\left(j\omega L_r+\frac{1}{j\omega C_r}+R_r+R_{eq}\right)i_r+j\omega M i_t=0 \tag{2.3}$$

Given that a diode rectifier load is employed in this study, the load resistance, R_{eq} can be expressed as:

$$R_{eq}=\frac{8R_{Load}}{\pi^2} \tag{2.4}$$

Where R_{Load} is the load of DC-link.

Based on the preceding formulations, the input impedance at the transmitter side can be derived as follows:

$$Z_{in}=\left(j\omega\, L_t+\frac{1}{j\omega C_t}+R_t\right)+\frac{\omega^2M^2}{j\omega\, L_r+\frac{1}{j\omega C_r}+R_r+R_{eq}} \tag{2.5}$$

An optimum load resistance exists for the wireless power transfer system to achieve maximum efficiency, and it can be derived as follows [11]:

$$R_{eq_{opt}}=R_r\left(\sqrt{1+\frac{\omega^2M^2}{R_tR_r}}\right) \tag{2.6}$$

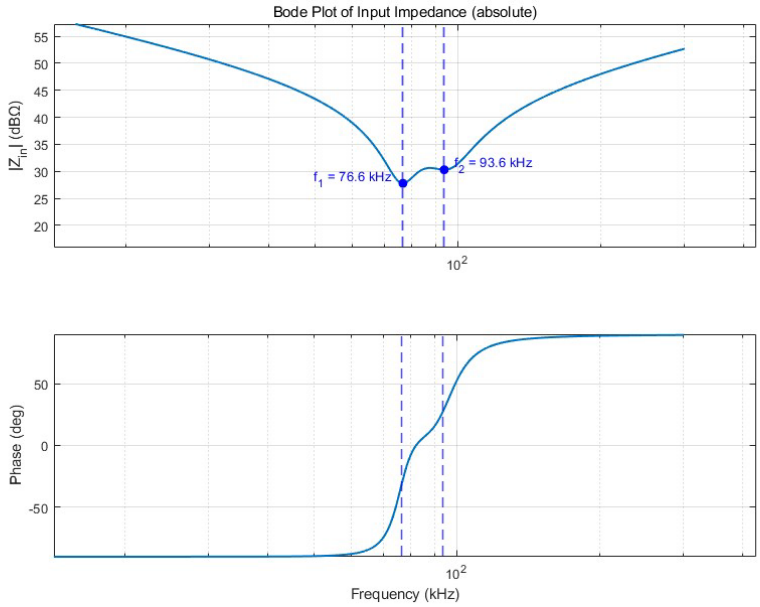

Fig 2.3 Bode plot of input impedance Z_{in}

As illustrated in Fig 2.3, the bode plot of the input impedance for the WPT system reveals a substantial increase in the magnitude of Z_{in} at frequencies beyond f_2 or below f_1. This behavior indicates that the series-series compensated WPT system exhibits bandpass filter characteristics. Furthermore, Z_{in} demonstrates inductive properties when the switching frequency is slightly above the resonant frequency f_r, and capacitive properties when the switching frequency is slightly below f_r.

2.2 Advanced-SVPFM method

Fig 2.1 shows the figure of A-SVPFM WPT system, the control part of the proposed three-level converter consists of the A-SVPFM module and the flying-capacitor voltage balancing method.

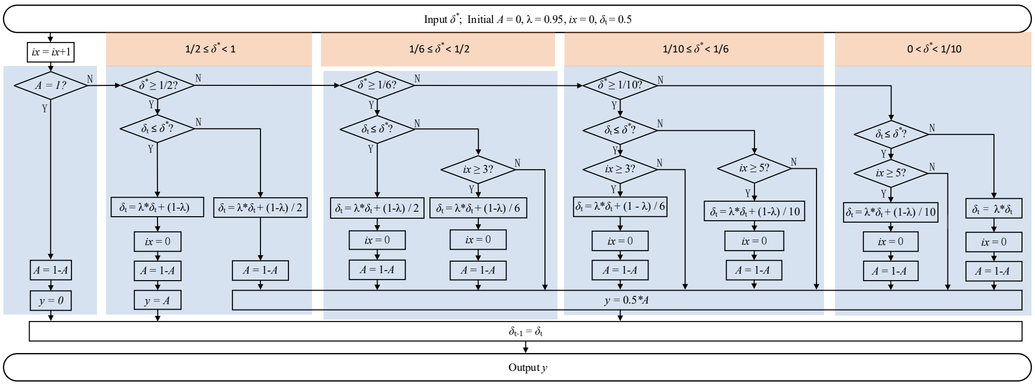

Fig.2.4 The proposed A-SVPFM in single control cycle

Fig 2.4 shows the control cycle of the proposed A-SVPFM method, which is half of the resonant cycle of the proposed WPT system, and it comprises the following steps:

First, the given command value of normalized voltage ratio for A-SVPFM δ^* in the range of (0,1), is read as input. The A-SVPFM method generates switching signals based on the given δ^*, and selects the appropriate $S_1 \sim S_4$ switching states through the voltage equalization algorithm discussed in later chapter.

Next, the variable A with the initial value 0 is conditionally toggled depending on δ^* and internal state variables as the normalized voltage ratio δ_t and the counter for control cycles ix. The equation for A toggling in the A-SVPFM as in Fig 2.4:

$$A_{t+1} = 1 - A_t \tag{2.7}$$

While A toggles in every control cycle for higher output levels when $\delta^* \geq 0.5$, it intentionally remains constant over multiple cycles for lower δ^* values, enabling reduced pulse frequency and ensuring soft-switching. Therefore, different to the BBPMM method from [30], which use variable A to ensure a 0 level be output for every control cycle. As shown in Fig 2.4, in the A-SVPFM method, A does not toggle strictly every control cycle, but rather follows a state-dependent toggling logic.

The root mean square (RMS) value of the fundamental component of the output voltage of the converter can be defined as follows:

$$U_{RMS} = \delta \cdot \frac{\sqrt{2}V_{dc}}{\pi} \tag{2.8}$$

where δ is the normalized voltage ratio; V_{dc} is the DC-side voltage of the converter.

The normalized voltage ratio δ defined in A-SVPFM is similar to the duty ratio defined in the PWM, by adjusting δ, the fundamental amplitude of the output voltage can be linearly controlled, thereby controlling the transmission power.

The PFM uses fixed-magnitude voltage pulses at adjacent frequencies to construct the desired output power. Thus, as shown in Fig 2.4, δ^* is divided into four intervals: $(0, 1/10)$, $[1/10, 1/6)$, $[1/6, 1/2)$, $[1/2, 1)$.

When $1/n \leq \delta_t < 1/m$ and $\delta_t \leq \delta^*$, where n and m are boundary values in the above four intervals, respectively, the normalized voltage ratio δ_t is calculated as:

$$\delta_t = \lambda\delta_{t-1} + (1 - \lambda)/\mathrm{m} \tag{2.9}$$

When $1/n \leq u_t < 1/m$ and $\delta_t > \delta^*$, the normalized voltage ratio δ_t is calculated as:

$$\delta_t = \lambda\delta_{t-1} + (1 - \lambda)/\mathrm{n} \tag{2.10}$$

where δ_{t-1} is the normalized voltage ratio of the previous control cycle, δ^* is the command value of the normalized voltage ratio for the A-SVPFM.

It is noticed that δ_{t0} must belong to (0, 1) to ensure the proper functioning of the proposed A-SVPFM method.

Furthermore, the basic principle of the proposed A-SVPFM will be explained clearly. In particular, it needs to be clarified how δ^* in A-SVPFM is proportional to the rms value of the fundamental component of the voltage output of the multilevel converters.

According to (9) and (10), while $0 < \delta^* < 1$, δ_t gradually increases or decreases to δ^*, then it remains in an interval. When $1/n \leq \delta^* < 1/m$, δ_t in the steady state is derived as:

$$\lambda\,\delta^* + \frac{1-\lambda}{m} \le \delta_t \le \lambda\delta^* + \frac{1-\lambda}{n} \tag{2.11}$$

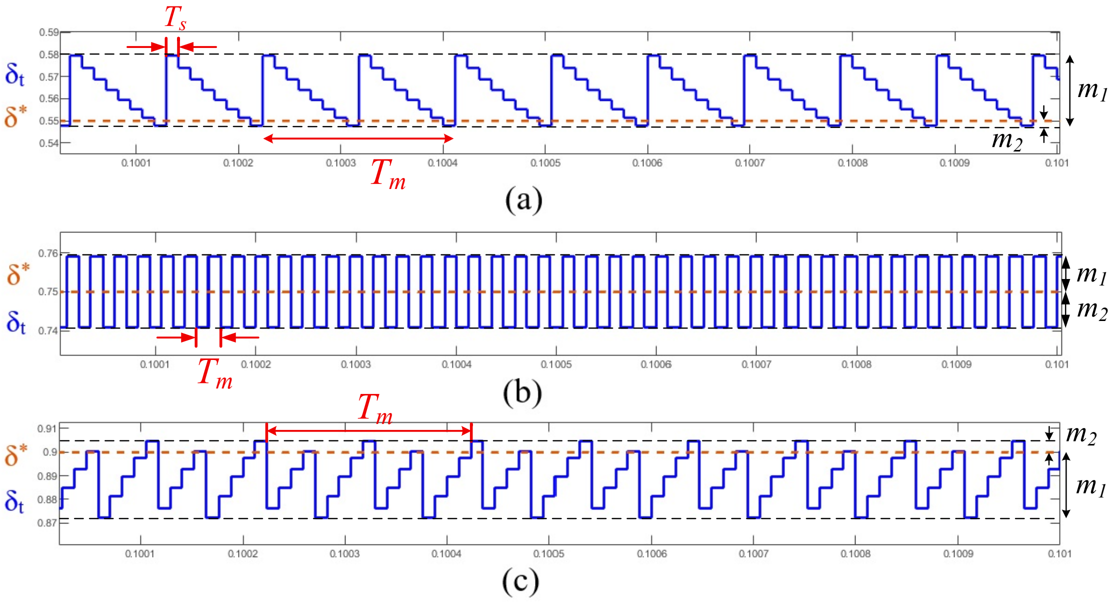

Fig 2.5 Time-domain wave of δ_t for different δ^* when $\lambda = 0.95$ (a) $\delta^* = 0.55$. (b) $\delta^* = 0.75$. (c) $\delta^* = 0.90$.

First, as shown in Fig 2.5, m_1 is defined as the difference between the maximum value of u_t and the command value δ^*, and m_2 is defined as the difference between δ^* and the minimum value of δ_t. When $1/n \le \delta^* < 1/m$, m_1 and m_2 can be calculated as follows:

$$m_1 = (1-\lambda)\left(\frac{1}{n} - \delta^*\right) \tag{2.12}$$

$$m_2 = (1-\lambda)\left(\delta^* - \frac{1}{m}\right) \tag{2.13}$$

According to (12) and (13), the width of the interval can be expressed as follows:

$$m_r = m_1 + m_2 = (1-\lambda)\left(\frac{1}{n} - \frac{1}{m}\right) \tag{2.14}$$

It can be found from (14) that λ determines the width of the interval, and a bigger λ will lead to a narrower width of the interval. The control object of the proposed A-SVPFM is the difference between m_1 and m_2, which is defined as the weight factor of the voltage pulse with the adjacent magnitudes of the output voltage of the multilevel converter:

$$\Delta m = m_1 - m_2 = (1-\lambda)\left(\frac{1}{n} + \frac{1}{m} - 2\delta^*\right) \tag{2.15}$$

In the range of $[1/n, 1/m)$, when $\delta^* = (m+n)/(2nm)$, $\Delta m = 0$, $m_1 = m_2$. In this case, it indicates that high/low amplitude pulses occur at the same frequency.

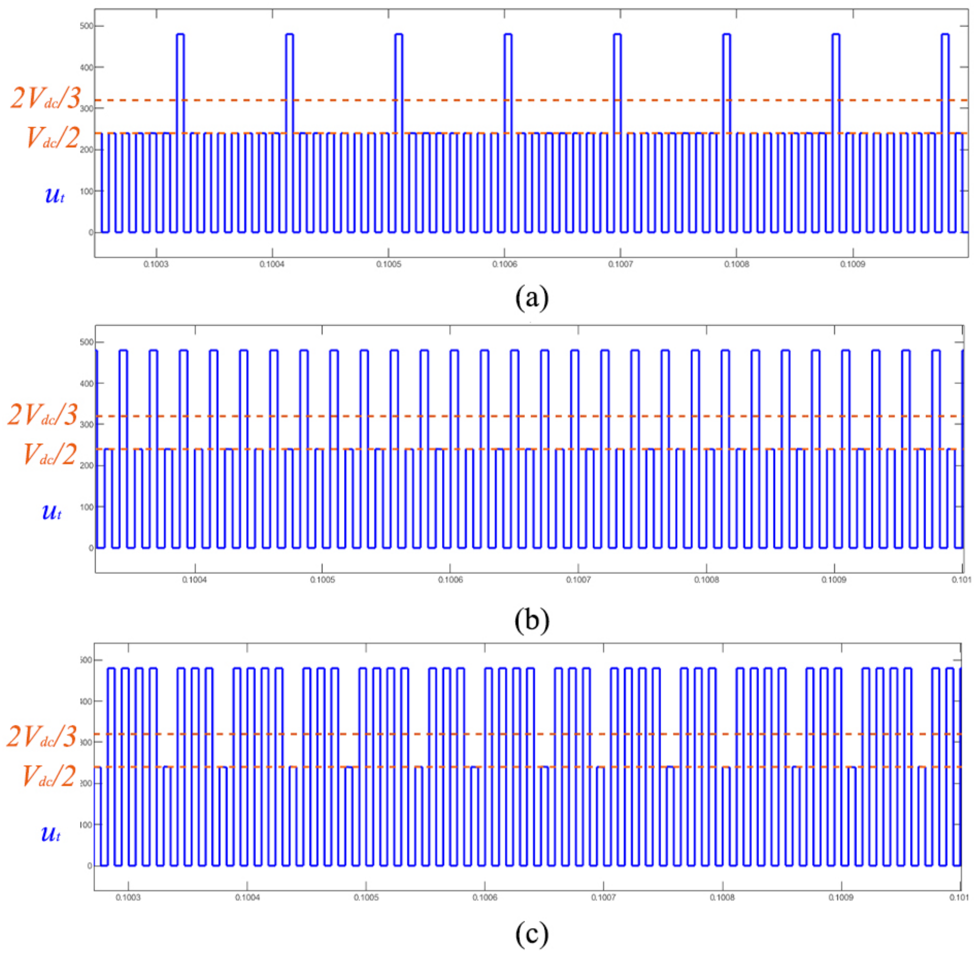

Fig 2.6 Time-domain wave of u_t for different δ^* when $\lambda = 0.95$ (a) $\delta^* = 0.55$. (b) $\delta^* = 0.75$. (c) $\delta^* = 0.90$.

For Fig 2.6, in this three-level half-bridge A-SVPFM WPT system, the output is only been switched in the adjacent amplitude set $\{0.5,1\}$; Define the whole wave voltage pulse count as N_1 and N_2 with δ^* in the interval of $[1/n, 1/m)$, then the normalized equivalent voltage ratio and modulation period are as:

$$\delta_{\text{A-SVPFM}} = \frac{\delta_1 \cdot N_1 + \delta_2 \cdot N_2}{N_1 + N_2} \tag{2.16}$$

Where δ_1 and δ_2 is the value of δ in the interval of $[1/n, 1/m)$.

To simplified explain (16), Table 2.1 shows the voltage vector selection method the three-level half-bridge converter in detail.

As shown in Table 2.1, while $0 \leq \delta^* < 1/10$, voltage vectors of $\delta_1 = 0$ and $\delta_2 = 1/10$ are combined to construct the desired δ^*; while $1/10 \leq \delta^* < 1/6$, voltage vectors of $\delta_1 = 1/10$ and $\delta_2 = 1/6$ are combined to construct the desired δ^*; while $1/6 \leq \delta^* < 1/2$, voltage vectors of $\delta_1 = 1/6$ and $\delta_2 = 1/2$ are combined to construct the desired δ^*; while $1/2 \leq \delta^* < 1$, voltage vectors of $\delta_1 = 1/2$ and $\delta_2 = 1$ are combined to construct the desired δ^*.

$\delta_1 = 0$	$\delta_1 = 1/10$	$\delta_1 = 1/6$	$\delta_1 = 1/2$
$\delta_2 = 1/10$	$\delta_2 = 1/6$	$\delta_2 = 1/2$	$\delta_2 = 1$
$\frac{1/10 \cdot N_2}{N_1 + N_2}$	$\frac{1/10 \cdot N_1 + 1/6 \cdot N_2}{N_1 + N_2}$	$\frac{1/6 \cdot N_1 + 1/2 \cdot N_2}{N_1 + N_2}$	$\frac{1/2 \cdot N_1 + N_2}{N_1 + N_2}$
$0 \leq \delta^* < 1/10$	$1/10 \leq \delta^* < 1/6$	$1/6 \leq \delta^* < 1/2$	$1/2 \leq \delta^* < 1$

Table 2.1: Combination of different ${\delta^*}_{\text{A-SVPFM}}$

When N_1 is coprime to N_2, T_m takes the minimum solution. According to (15), it can be seen that $\delta_{I-SVPFM}$ is completely determined by the density ratio of the two amplitude pulses. The modulation period T_m can be expressed as:

$$T_m = (N_1 + N_2)T_s, \qquad T_s = \frac{1}{f_r} \tag{2.17}$$

Where T_s is the resonance half-period, f_r is the resonance frequency,

In this three-level half-bridge A-SVPFM WPT system, the resolution is mainly determined by the factor λ, when λ increases, the swing amplitude of u_t around the instruction u becomes narrower, and the output equivalent value is closer to the instruction, so the resolution is advanced and the deviation is reduced; intuitively, Fig. 5 presents a monotonic relationship of "λ increases = deviation decreases, resolution increases". When λ is high (e.g. λ≈0.93), the resolution of this method can be equal to or even better than Δ-Σ PFM, both of which can reach the order of about 10^{-4}.

2.3 Harmonic Analysis of A-SVPFM and PFM

The output voltage $u_t(t)$ of the multilevel converter is expanded as a Fourier series as:

$$u_t\ (t) = \sum_{-\infty}^{\infty} k_n e^{2in\pi f_0 t} \tag{2.18}$$

where f_0 is called the fundamental frequency, and k_n is the Fourier coefficient at the nth harmonic. The harmonic frequency can be derived as follows:

$$f_h = \left(1 \pm \frac{h}{N_t}\right)(2n-1)f_0, h = 0,1,2 \ldots., N_{t-1} \tag{2.19}$$

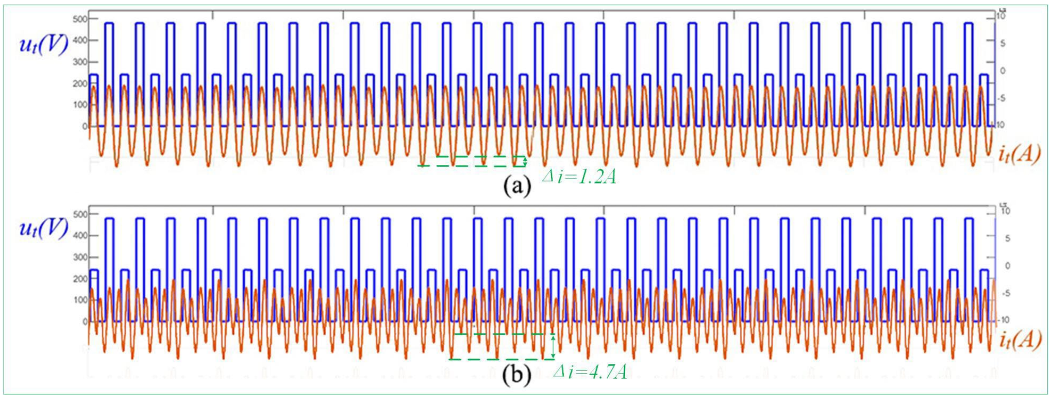

Fig 2.7 Time-domain waves comparison of u_t and i_t when $\lambda = 0.95$, $\delta^* = 0.75$. (a) A-SVPFM. (b) traditional Δ-Σ PFM.

Since a full modulation period T_m encompasses multiple resonant cycles, the output voltage spectrum ut exhibits both subharmonic and interharmonic components. A key distinction between A-SVPFM and Δ-Σ PFM lies in the interharmonics characteristics of the output voltage. Figure 2.7 provides a comparative analysis of the time-domain waveforms of u_t and the transmitter current i_t in the WPT system under A-SVPFM and Δ-Σ PFM modulation methods.

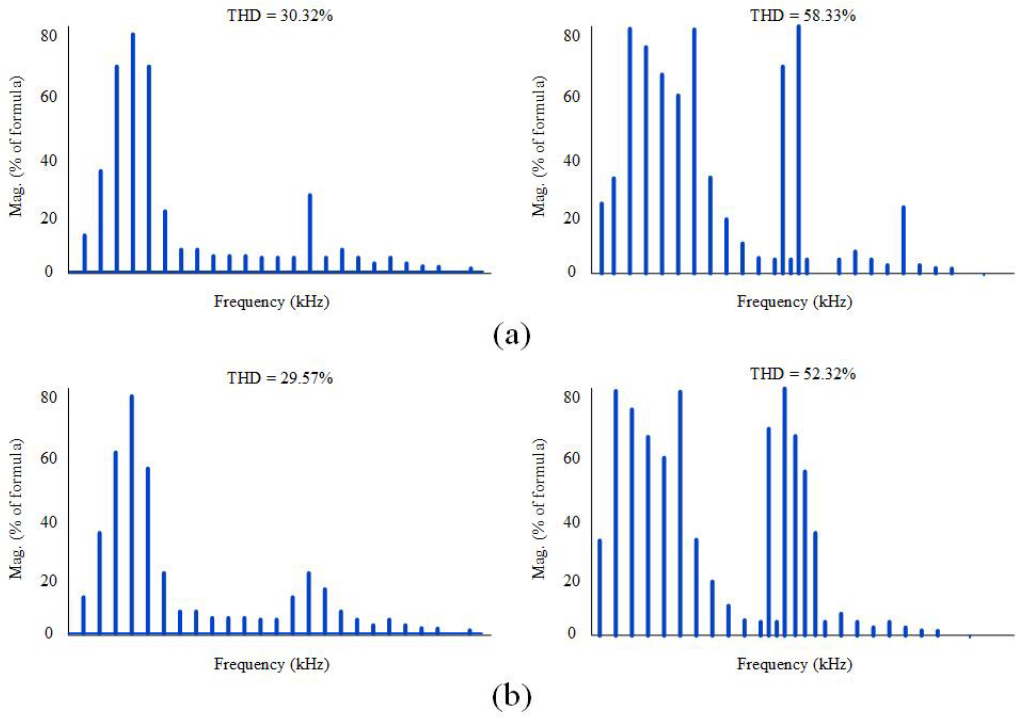

Fig 2.8 Comparison of DFT of u_t and i_t when $\lambda = 0.95$, $\delta^* = 0.75$. (a) A-SVPFM. (b) traditional Δ-Σ PFM.

The principal distinction between A-SVPFM and Δ-Σ PFM lies in their respective interharmonics profiles. Owing to the impedance properties of the WPT system, interharmonics frequencies near the resonant frequency are particularly influential. The expression for the interharmonics frequencies, given in Eq. (19), shows their dependence on the modulation period T_m. Although the amplitude of these interharmonics is influenced by the voltage pulse sequence and is challenging to derive analytically, it can be effectively evaluated using discrete Fourier transform (DFT). Figure 2.8 presents a comparative DFT analysis of u_t and i_t under A-SVPFM and Δ-Σ PFM.

The results from Fig 2.7 and Fig 2.8 demonstrate that the total harmonic distortion (THD) of the transmitter current in the A-SVPFM-based WPT system is significantly lower compared to using Δ-Σ PFM. This phenomenon can be attributed to the reduced modulation period resulting from adjustments to the A-SVPFM resolution, which in turn generates distinct interharmonics components near the resonant frequency, as shown in Figures 2.7.

Chapter 3

Capacitor Voltage Balancing of WPT system

3.1 Flying-capacitor Voltage Balancing method

Maintaining the flying-capacitor voltage balance in the WPT system is critical, and this balance is primarily governed by the switching modes of the half-bridge converter.

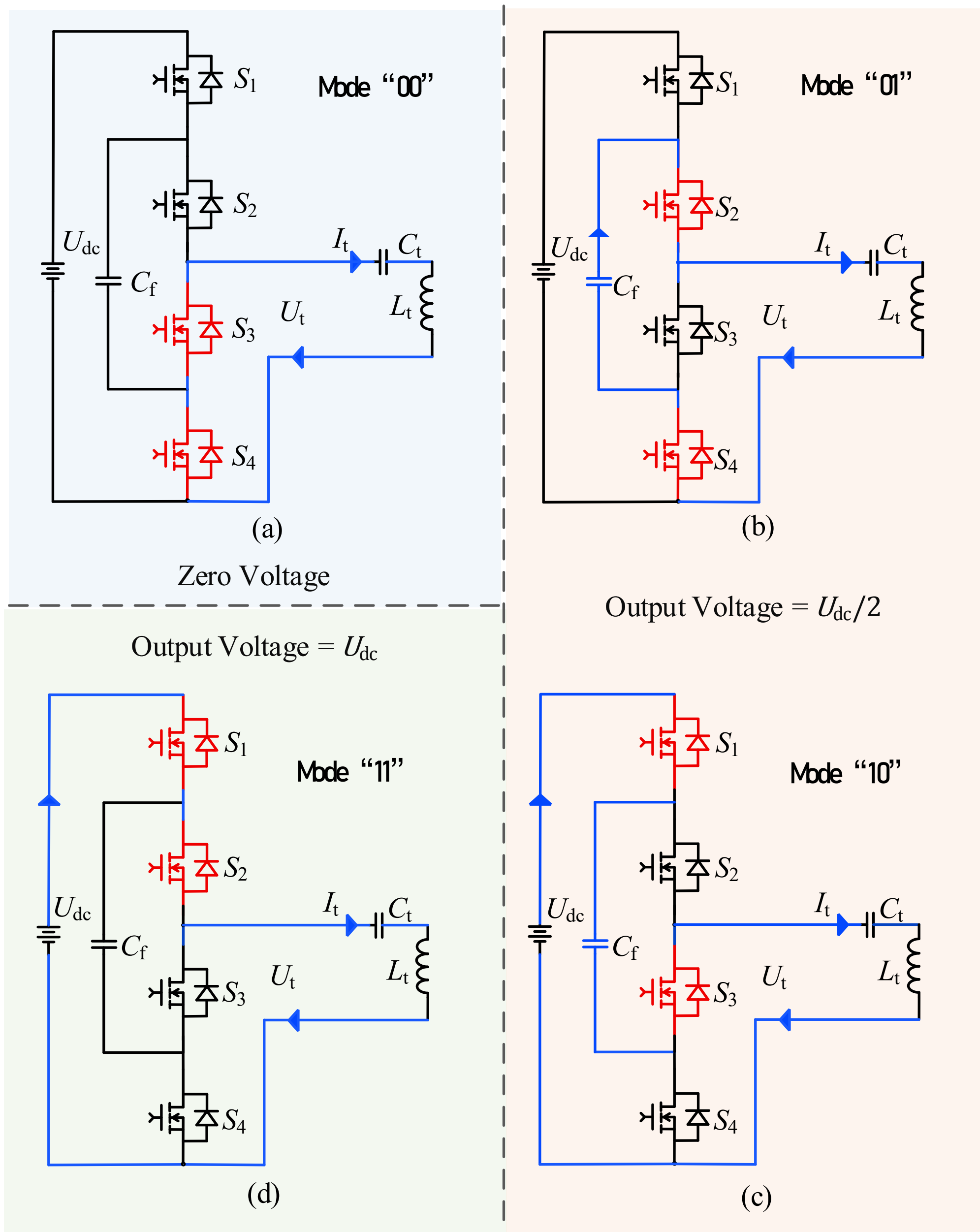

Fig 3.1 Switching modes of three-level half-bridge converter. (a) State "00", zero voltage. (b) State "01", $U_{dc}/2$, capacitor discharging. (c) State "10", $U_{dc}/2$, capacitor charging. (d) State "11", U_{dc}.

Fig. 3.1 shows the 4 valid switching modes of the proposed three-level half-bridge converter.

Figure 3.1(a) depicts the "00" operating mode of the flying-capacitor three-level converter, in which S1 and S2 are turned OFF while S3 and S4 are turned ON, yielding zero output voltage and maintaining the flying-capacitor voltage.

Figure 3.1(b) depicts the "01" operating mode, in which S1 and S3 are turned

OFF while S2 and S4 are turned ON.

Figure 3.1(c) depicts the “10” operating mode, in which S2 and S4 are turned OFF while S1 and S3 are turned ON, and in both “01” and “10” modes the output voltage equals $U_{dc}/2$.

Figure 3.1(d) depicts the “11” operating mode, in which S3 and S4 are turned OFF while S1 and S2 are turned ON, producing an output voltage of U_{dc} while preserving the flying-capacitor voltage.

When the transmitter current is positive, mode “01” charges the flying capacitor, whereas mode “10” discharges it; conversely, when the transmitter current is negative, mode “01” discharges the capacitor and mode “10” charges it. Consequently, modes “10” and “01” have the function of regulate and balance the flying capacitor voltage.

As discussed earlier, the voltage levels U_{dc} and 0 each correspond to a single switching configuration—achieved by turning ON all upper switches or all lower switches respectively. Because the flying capacitor is not engaged in these configurations, the states associated with U_{dc} and 0 do not affect capacitor voltage balancing. In contrast, the 0.5 U_{dc} level admits redundant switching states; therefore, two states that induce opposite charging directions for the flying capacitor should be selected.

A three-level half-bridge converter can generate three distinct output voltage levels. Because the transmitter current i_t is negative when the converter output voltage decreases and positive when the converter output voltage increases, the transmitter current i_t can be inferred from the switching state without measurement.

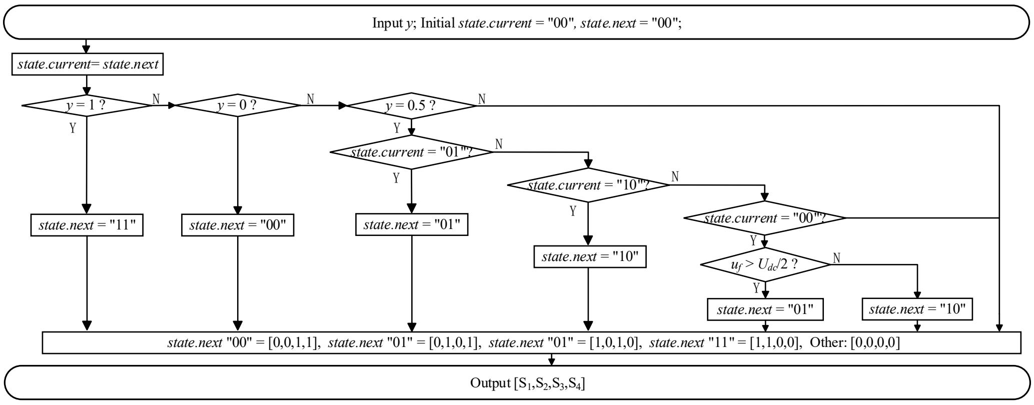

Fig 3.2 Flow chart of proposed voltage balancing method

Figure 3.2 is the flow chart of the proposed switch state selection and voltage balancing method. As shown in Figure 3.2, the flying capacitor voltage (u_f) needs to be measured. Record the last switch state in each control cycle, denoted as "$state.current$" in the figure.

In the voltage balancing process, if the output of A-SVPFM equals 1, i.e. the input of voltage balancing method $y = 1$, the switch state is selected to be "11". If $y = 0$, the switch state is selected as "00".

When $y = 0.5$, if "$state.current$" is “01”, “$state.next$” should be set to “01”; or if "$state.current$" is “10,” then set “$state.next$” to “10”. This is because of considering that there are three possibilities of voltage level U_{dc}, $U_{dc}/2$ or 0, if the last switching state is "10" or "01", the switching state should be maintained. Therefore, in the switching state from "10" to "10" or "01" to "01", so that two charge and discharge process can cancel each out.

Otherwise, for the purpose of balancing flying-capacitor voltage, it is necessary to judge the state "10" or "01" when "$state.current$" is “00”. If the last voltage level is 0, i_t will be positive during that switching cycle. In this case, if u_f is greater than the reference value (i.e., $U_{dc}/2$), switch state "01" should be selected to discharge the capacitor; else, switch state "01" should be selected to charge the capacitor.

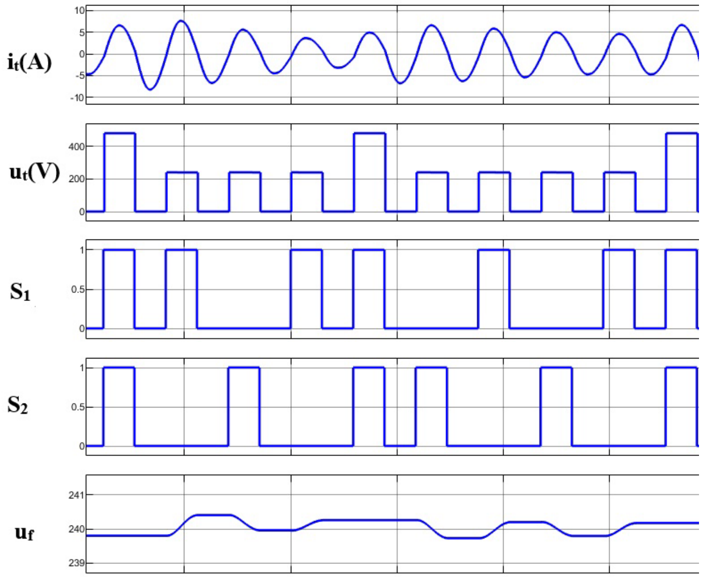

Fig 3.3 Waveform of the flying capacitor charging/discharging when $\delta = 0.6$

Fig 3.3 depicts an example of the operational cycle of the proposed charging and discharging mechanism under the condition where the given voltage-ratio $\delta = 0.6$. As observed, only a unidirectional half-wave component of the transmitter current is utilized for energy transfer. The charging and discharging processes of the flying capacitor in the three-level half-bridge converter are governed by the time integral of the transmitter current within the WPT system. The balancing algorithm operates at twice the switching frequency to minimize voltage fluctuations across the flying capacitor.

The resonant transmitter current is approximated by a sinusoidal waveform:

$$i_t(t) = I_t \sin(\omega_0 t) \tag{3.1}$$

Where ω_0 is the angular resonant frequency. Of the period, the smallest positive T satisfying $i_t(t+T) = i_t(t)$ must fulfill:

$$\omega_0 t = 2\pi \tag{3.2}$$

So that:

$$T_s = \frac{2\pi}{\omega_0} \tag{3.3}$$

$$\Delta u_{\mathrm{f}} = \frac{1}{C_{\mathrm{f}}}\int_0^{\frac{T_s}{2}} i_{\mathrm{t}}(t)dt \approx \frac{1}{C_{\mathrm{f}}}\int_0^{\frac{\pi}{\omega_0}} I_{\mathrm{t}} \sin(\omega_0 t)\, dt = \frac{2I_{\mathrm{t}}}{\omega_0 C_{\mathrm{f}}} \tag{3.4}$$

Accordingly, over one control cycle, the net change in flying-capacitor voltage can be derived as follows:

$$\Delta u_{\mathrm{f}} = \frac{I_{\mathrm{t}} T_{\mathrm{s}}}{\pi C_{\mathrm{f}}} \tag{3.5}$$

where I_t is the peak value of the transmitter current, C_f is the capacity of flying capacitor, T_s is the control cycle, in a control cycle, the capacitor is charged or discharged only in half a cycle, so the integration interval is $T_s/2$.

When the given voltage-ratio $\delta = 1$, the fundamental component of it for the WPT system is maximum. In this case, the average transmitter current is twice the dc-side current:

$$I_{\mathrm{tr}} = \frac{\pi P_{\mathrm{r}}}{U_{\mathrm{dc}}} \tag{3.6}$$

where I_{tr} is the rating value of the I_t; P_r is the rating input power of the WPT system.

Ideally, $I_t = I_{tr}$ considering the influence of harmonics, the peak value of actual emission current will be 0%~20% larger than the rated fundamental component, thus the maximum peak value of the transmitter current can be given as:

$$I_{\mathrm{tmax}} = kI_{\mathrm{tr}}, \; k \in (1,1.2) \tag{3.7}$$

where k is a correction factor.

According to (3.1)– (3.7), the voltage ripple of the flying capacitor can be expressed as follows:

$$\Delta u_{\mathrm{f}} = \frac{1}{C_{\mathrm{f}}} \int_0^{\frac{T_{\mathrm{s}}}{2}} i_{\mathrm{t}}(t) dt = \frac{I_{\mathrm{t}} T_{\mathrm{s}}}{\pi C_{\mathrm{f}}} \le \frac{k P_{\mathrm{r}}}{C_{\mathrm{f}} U_{\mathrm{dc}} f_{\mathrm{s}}} \tag{3.8}$$

As shown in Fig.3.3, the voltage ripple is consistent with (3.8). Based on (3.8), the capacitor should be selected as follows:

$$C_{\mathrm{f}} \ge \frac{2kP_{\mathrm{r}}}{\varepsilon {U_{\mathrm{dc}}}^2 f_{\mathrm{s}}} \tag{3.9}$$

where ε is the voltage ripple rate of the flying capacitor, and $\varepsilon = 2\Delta u_f/U_{dc}$.

3.2 Zero Voltage Switching Analysis

According to (2.18), the Fourier series expansion of the output voltage waves (u_t) of the multilevel converter using A-SVPFM can be further derived as:

$$u_{\mathrm{t}}(t) = U_{\mathrm{dc}} \cdot (\frac{\delta_{\mathrm{A-SVPFM}}}{2} + \delta_{\mathrm{A-SVPFM}} \sum_{n=1}^{\infty} a_k(\delta) \cdot \sin(2k+1)\omega t) \quad (3.10)$$

where $a_k(\delta) = \frac{2}{(2k+1)\pi}$ denote the odd-harmonic amplitude coefficients respectively.

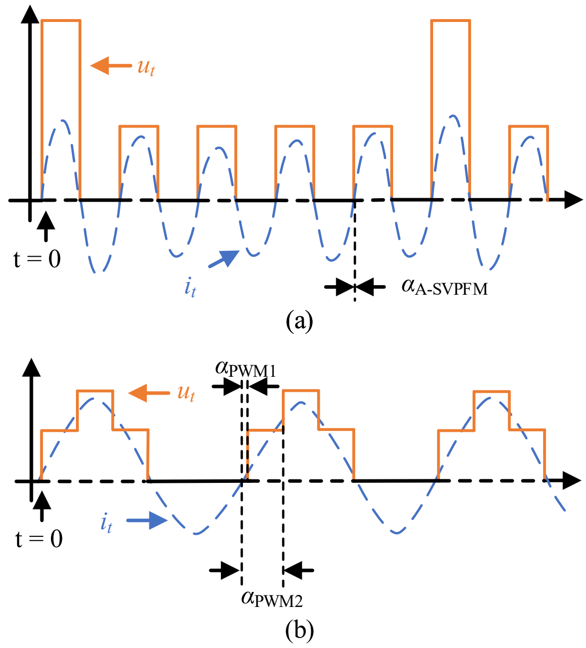

Fig 3.4 Time-domain waves of three-level converter by using (a) A-SVPFM, $\delta_{\mathrm{A-SVPFM}} = 0.6$ and (b) PWM, $d_{pwm} = 0.6$.

According to (3.10), the switching angle of A-SVPFM $\propto_{I-SVPFM}$ as shown in Fig. 3.4 (a)is always equal to zero. While the switching frequency is equal to the resonant frequency, the phases of u_t and it are consistent. By comparison, selecting $t = 0$ as the timeline, u_t of the multilevel converter using the conventional PWM can be derived as [26]:

$$u_t = \frac{(D_1 + D_2)U_{\mathrm{dc}}}{2} + \sum_{n=1}^{\infty} \frac{U_{\mathrm{dc}}}{n\pi} (\sin n\pi D_1 + \sin n\pi D_2) \cos[n\pi D_1 - n\omega t] \tag{3.11}$$

From article [26], the switching angle of PWM ($\propto_{PWM1}$) as shown in Fig 3.4(b) is derived as

$$\propto_{PWM1}, \ n = (0.5 - D_1)n\pi \tag{3.12}$$

Similarly, selecting $t1 = 0$ as the timeline, $\propto_{PWM2}$ is derived as

$$\propto_{PWM2}, \ n = (0.5 - D_2)n\pi \tag{3.13}$$

Consequently, for conventional PWM schemes, a zero-switching angle is attainable solely under the condition that $D_1 = D_2 = 0.5$. As the duty ratios deviate from this equilibrium and assume smaller values, the resultant switching angles increase in magnitude. This phenomenon indicates that zero-voltage switching cannot be sustained when adjusting the PWM duty cycle. In contrast, the A-SVPFM technique maintains a consistent zero switching angle across variations in the voltage ratio $\delta_{I-SVPFM}$, thereby facilitating more robust and straightforward achievement of ZVS compared to PWM-based methods.

The input impedance of a S-S compensated WPT system exhibits inductive characteristics when the system operates at a switching frequency marginally exceeding the resonant frequency, or when the transmitter capacitance is augmented. In the case of an increase in transmitter capacitance, the input impedance can be reformulated as follows:

$$Z_{in}(\omega_r, \Delta C) \approx R_{eq,t} + j \cdot \frac{\Delta C}{\omega_r C_t (C_t - \Delta C)} \tag{3.14}$$

Where:

$$R_{eq,t} \triangleq R_t + \frac{{\omega_r}^2 M^2}{R_r + R_{eq}} \tag{3.15}$$

According to (3.14) and (3.15), the magnitude and phase of Z_{in} at the resonant frequency can be approximately regarded as

$$|Z_{in}| \approx R_{eq,t} \tag{3.16}$$

$$\tan \propto \approx k_z \frac{\Delta C}{C_t}, \qquad k_z = \frac{1}{\omega_r C_t R_{eq,t}} \tag{3.17}$$

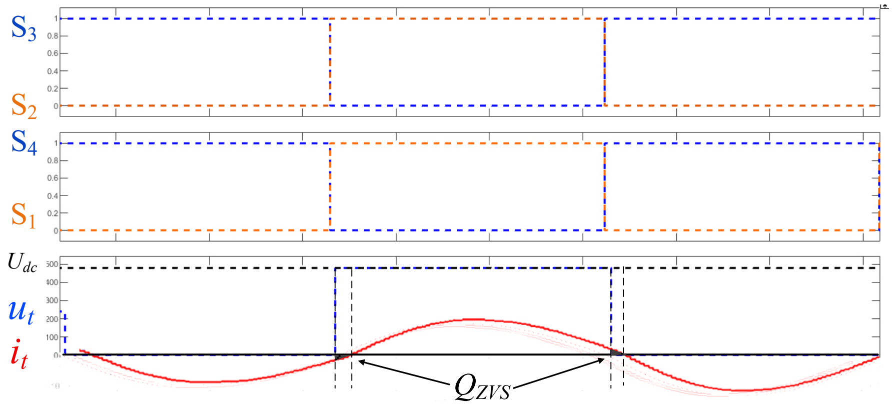

Fig 3.5 Operating waveforms of switching state change from "00" to "11".

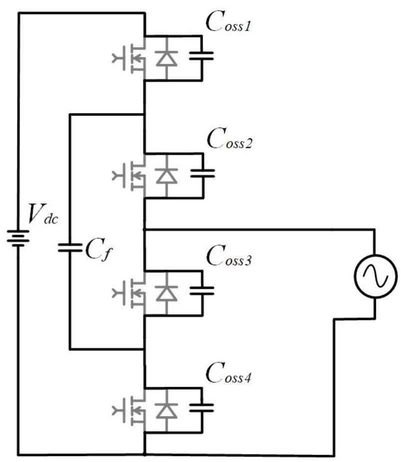

Fig 3.6 Equivalent circuit of three-level converter during dead time.

During the transition from switching state "00" to "11" as in Fig 3.5, the transmitter current i_t exhibits a slight phase lag relative to the output voltage u_t throughout the dead time, corresponding to a phase difference of α. And as illustrated in Fig 3.6, the transmitter current i_t discharges the parasitic output capacitances (C_{oss}) of S_1 and S_2, while simultaneously charging those of S_3 and S_4. Here, Q_{req} is the minimum charge required to charge or discharge the output capacitance of the associated power switch from 0 to $U_{dc}/2$ during a switching transition.

$$Q_{ZVS} = \int_0^{T_d} i_t \, dt \tag{3.17}$$

Q_{ZVS} represents the integral of i_t over the dead time interval.

$$Q_{req} = \int_{0}^{\frac{U_{dc}}{2}} (C_{OSS1} + C_{OSS2} + C_{OSS3} + C_{OSS4}) dv \tag{3.18}$$

Zero-voltage switching (ZVS) is achieved provided that Q_{ZVS} is sufficient to complete the charge or discharge of the relevant C_{OSS} capacitances:

$$Q_{ZVS} \geq Q_{req} \tag{3.19}$$

where C_{OSS1}, C_{OSS2}, C_{OSS3}, and C_{OSS4} represent the drain-source capacitors of S_1, S_2, S_3, and S_4, respectively.

Because the three-level half-bridge needs to charge and discharge the C_{OSS} of two upper/two lower tubes at the same time in this phase change, the voltage span is $U_{dc}/2$, so the right integral is the sum of four C_{OSS}.

Under the sinusoidal approximation of the transmitter current i_t during the dead time interval, the minimum required dead time T_{omin} can be derived from (3.17) - (3.19) and Fig 3.6 as follows:

$$T_{dmin} = \frac{1}{\omega_r} \cos^{-1}\left(1 - \frac{\omega_r U_{dc}(C_{OSS1} + C_{OSS2} + C_{OSS3} + C_{OSS4})}{I_t}\right) \tag{3.20}$$

According to (3.10), ignoring the influence of the harmonic current, I_t can be calculated as

$$I_t = \frac{2\delta_{\mathrm{A-SVPFM}} U_{dc}}{\pi R_{\mathrm{eq,t}}} \tag{3.22}$$

According to (3.15) - (3.22), the relationship between the desired increased capacitor (ΔC) and the load condition (R_{eq}) to realize the ZVS can be derived as the relationship of ΔC and $R_{\mathrm{eq,t}}$ as follows:

$$\frac{\Delta C}{C_t} \geq \sqrt{\frac{\pi \cdot (\omega_{\mathrm{r}} \cdot R_{\mathrm{eq,t}})^3 \cdot \sum C_{\mathrm{OSS}}}{2\delta_{\mathrm{I-SVPFM}}}} \tag{3.23}$$

Where $\sum C_{\mathrm{OSS}} = C_{OSS1} + C_{OSS2} + C_{OSS3} + C_{OSS4}$.

Chapter 4

Experimental Verification

4.1 System topology

To verify the feasibility of the proposed A-SVPFM-based WPT system, a practical system with a prototype is constructed for experimentation.

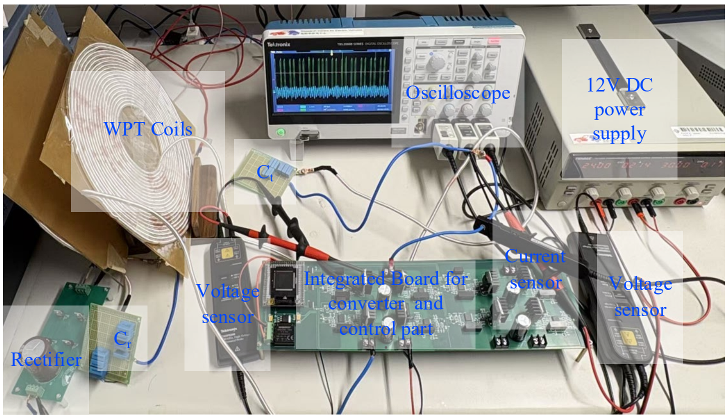

Fig 4.1 The experimental system.

As illustrated in Fig. 4.1, the setup comprises a dc power source, a power analyzer, a Gallium-Nitride-based three-level flying-capacitor half-bridge converter with drivers, a WPT circuit, a bridge rectifier, an electronic load, a 12V auxiliary supply, and a TMS320FDSP28377 control board. The transmitter and receiver coils are wound from copper Litz wire. The power analyzer is utilized to measure the input and output power.

The high-voltage DC power source (MR50040) provides the dc-link input voltage. The system parameters are listed in Table 4.1 as:

Items	Value
DC input voltage (U_{dc})	150V
Primary compensated capacitance (C_t)	31.17nF
Primary coil inductance (L_t)	83.6μH
Parasitic resistance of L_t in series with C_t	0.25Ω
Secondary compensated capacitance (C_r)	29.36 nF
Secondary coil inductance (L_r)	82.9 μH
Parasitic resistance of L_r in series with C_r	0.24Ω
Distance between two coils (d)	10cm
Mutual inductance (M)	64.2 μH
Resonant frequency (f_r)	99.8kHz
Flying capacitance (C_f)	22 μF
Load resistance (R_{load})	82.3 Ω
Dead time (T_d)	150 ns
Integration coefficient (K)	0.21
Digital signal processer	TMS320FDSP28377

Table 4.1: System parameters of experimental verification

4.2 Zero Voltage Switching

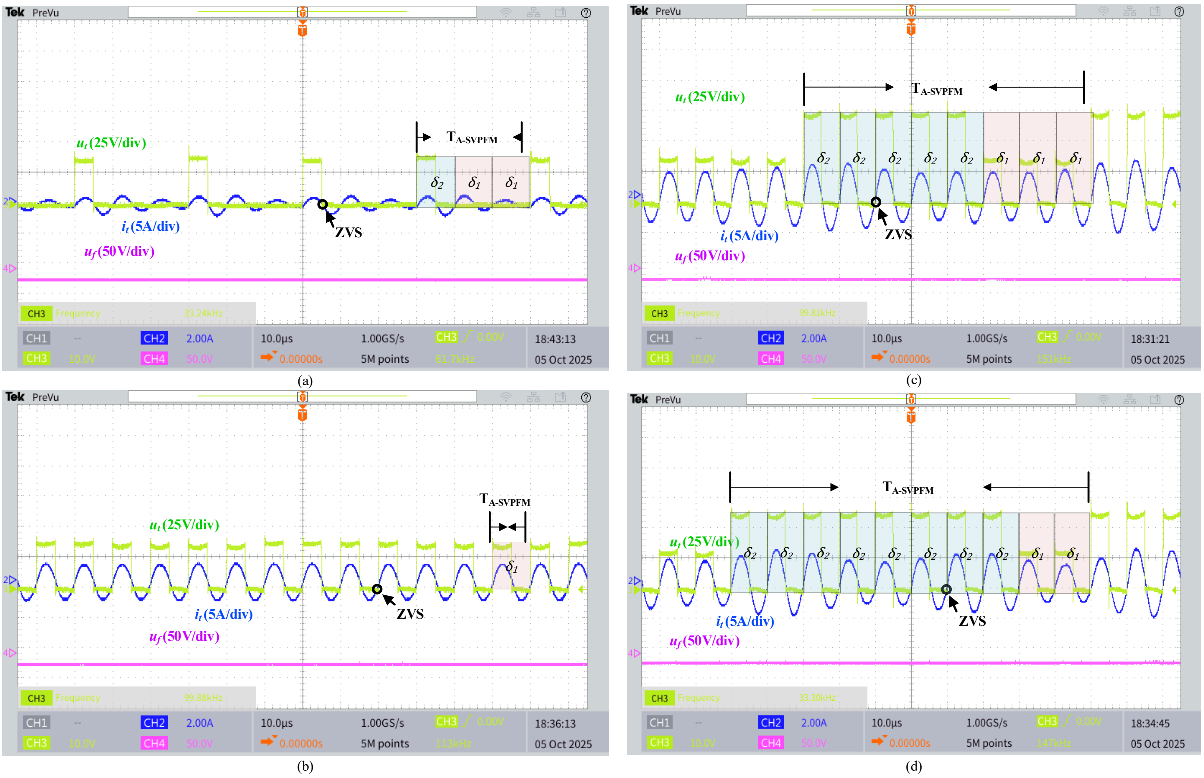

Fig 4.2 Measured waveforms of the WPT system using A-SVPFM (a) $\delta_{A-SVPFM}{}^{*} = 0.1$ (b) $\delta_{A-SVPFM}{}^{*} = 0.5$ (c) $\delta_{A-SVPFM}{}^{*} = 0.75$ (d) $\delta_{A-SVPFM}{}^{*} = 0.90$

As shown in Fig 4.5 the operating waveforms of the three-level converter-based WPT system by using the proposed A-SVPFM method.

When the given voltage-ratio $< \delta_{A-SVPFM}{}^{*} = 1$, the capacitor voltage u_c does not have to be balanced at the reference value. Otherwise, when $0 < \delta_{A-SVPFM} < 1$, Fig 4.4 displays that the capacitor voltage can be balanced at around half of the input voltage. Besides, the capacitor voltage ripples caused by the charging/discharging current are very small, because the capacitor is big enough, which matches closely with (22).

As shown in Fig 4.5(a), while $\delta_{A-SVPFM}{}^{*} = 0.1$, according to Table 2.1, so that:

$$\frac{\frac{1}{10} \cdot N_1 + \frac{1}{6} \cdot N_2}{N_1 + N_2} = 0.1 \tag{4.1}$$

The minimum solutions are $N_1 \approx 2$, $N_2 \approx 1$, respectively, to realize the minimum period of the output voltage. As shown in Fig. 14(a), u_t consists of one voltage pulses with $\delta = 1/6$ and zero vector.

As shown in Fig 4.5(b), while $\delta_{\mathrm{A-SVPFM}}{}^{*} = 0.5$, as shown in Table 2.1, the $\delta_{\mathrm{A-SVPFM}}{}^{*}$ belonging the condition of $1/2 \leq \delta_{\mathrm{A-SVPFM}}{}^{*} < 1$, so that:

$$\frac{\frac{1}{2} \cdot N_1 + N_2}{N_1 + N_2} = 0.5 \tag{4.2}$$

The minimum solutions of N_1 and N_2 can be calculated to 1 and 0, respectively, to realize the minimum period of the output voltage. As shown in Fig. 14(b), u_t consists of one voltage pulses with $\delta_1 = 1/2$ and zero voltage pulses with $\delta_2 = 1$ when $\delta_{\mathrm{A-SVPFM}}{}^{*} = 0.5$.

As shown in Fig 4.5(c), while $\delta_{\mathrm{A-SVPFM}}{}^{*} = 0.75$, the $\delta_{\mathrm{A-SVPFM}}{}^{*}$ belonging the condition of $1/2 \leq \delta_{\mathrm{A-SVPFM}}{}^{*} < 1$, so that:

$$\frac{\frac{1}{2} \cdot N_1 + N_2}{N_1 + N_2} = 0.75 \tag{4.3}$$

The minimum solutions of N_1 and N_2 can be calculated to be 1 and 1, respectively, to realize the minimum period of the output voltage. As shown in Fig. 14(c), u_t consists of two voltage pulses with $\delta_1 = 1/2$ and two voltage pulses with $\delta_2 = 1$ when $\delta_{\mathrm{A-SVPFM}} = 0.4$.

As shown in Fig 4.5 (d), $\delta_{\mathrm{A-SVPFM}}{}^{*} = 0.9$, belonging the condition of $1/2 \leq \delta_{\mathrm{A-SVPFM}}{}^{*} < 1$, as shown in Table 2.1:

$$\frac{\frac{1}{2} \cdot N_1 + N_2}{N_1 + N_2} = 0.9 \tag{4.4}$$

The minimum solutions can be calculated as $N_1 = 1$, $N_2 = 4$. Therefore, u_t consists of voltage pulses with the ratio of $\delta_1 = 1/2$ to $\delta_2 = 1$ as $1:4$.

The output voltage of the experiment as shown in Fig 4.4 (a)-(d) are consistent with both the calculated and simulation results, and with all range of δ^*,the experiment result demonstrate the desired ZVS. It means that the A-SVPFM enables the three-level converter to output voltage with the ideal solution of N_1 and N_2.

4.3 Step Response

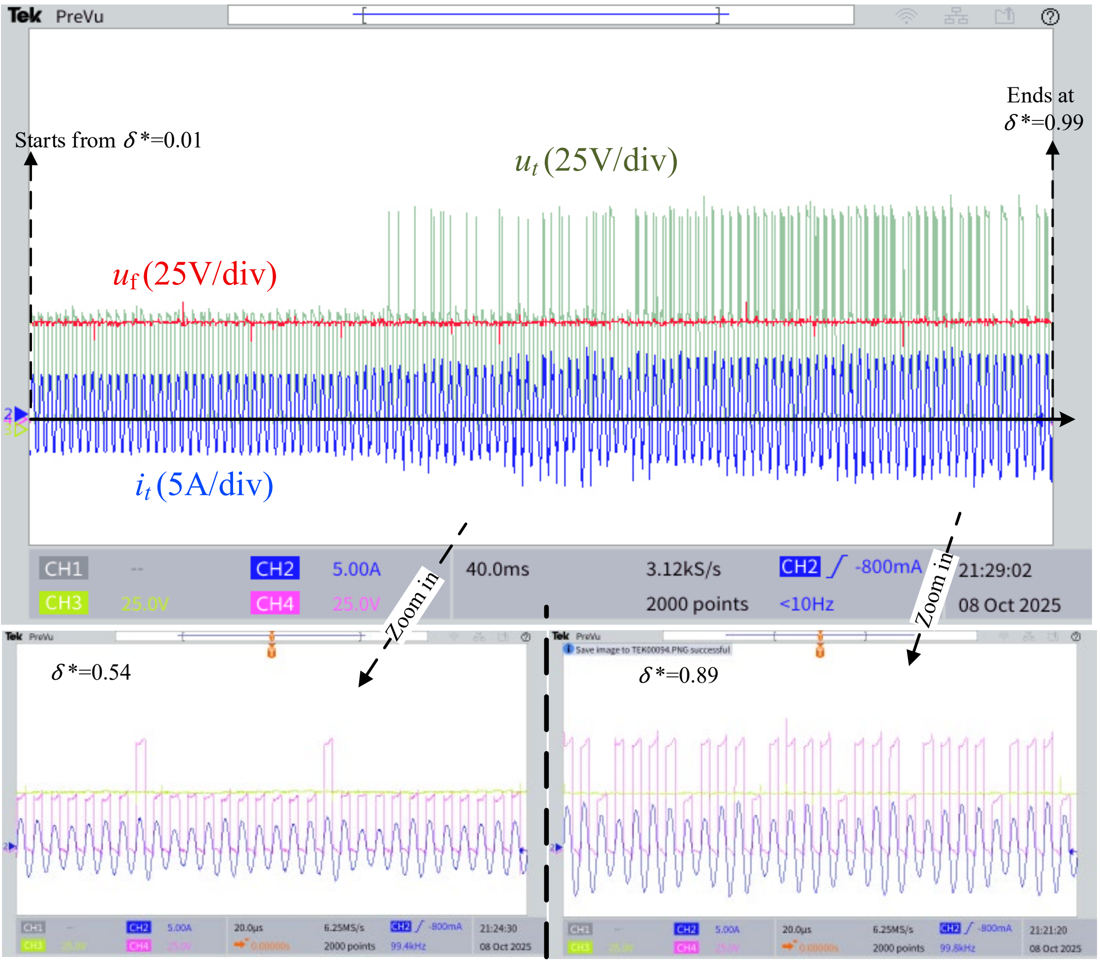

Fig 4.3 Step responses of proposed WPT system with increasing δ^*

Furthermore, as shown in Fig 4.6, under the proposed SVPFM with $\delta_{I-SVPFM}$ swept from 0.01 to 0.99, the step responses and their zoomed-in waveforms demonstrate that u_f remains essentially unchanged, while the WPT dc-side output voltage exhibits a rapid, overshoot-free step response.

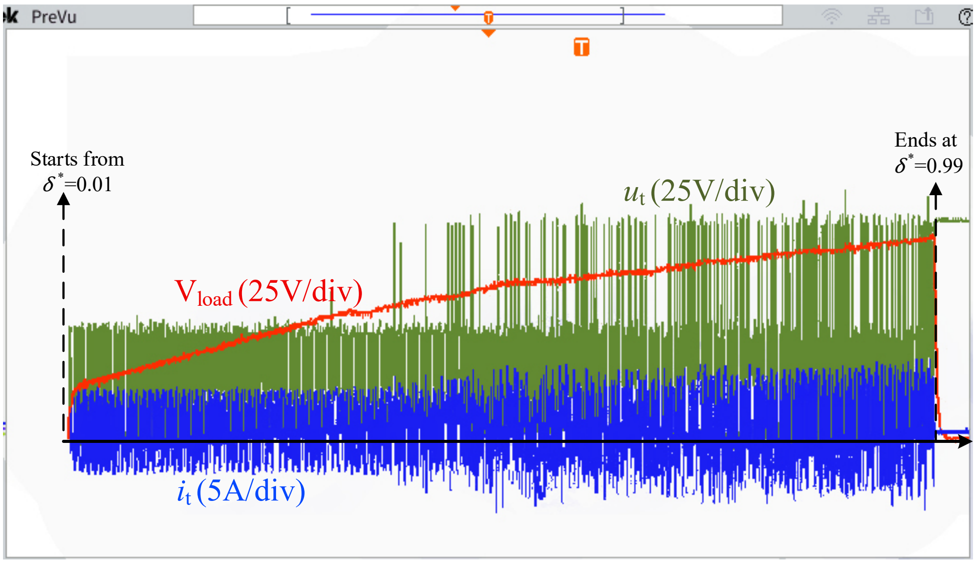

Fig 4.4 Output voltage of proposed WPT system with increasing δ^*

Considering the S-S compensated WPT system behaves the current-source characteristic, therefore, the load voltage of the receive side also has a fast step response without overshoot when δ^* increases. Figure 4.7 shows that WPT system using the proposed flying-capacitor voltage balancing method are stable for step response

4.4 System Efficiency

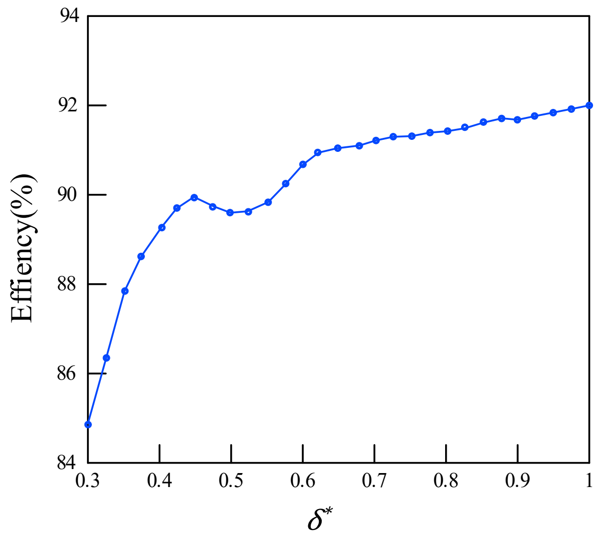

Fig 4.5 System efficiency of proposed WPT system with increasing δ^*

Fig. 17(a) shows the efficiency of the A-SVPFM WPT system with different output power by changing the δ^*. The maximum efficiency is measured as 91.84% when $\delta^* = 0.99$, where the output power is 215.6W.

Chapter 5

Conclusion

This dissertation proposed and implemented an advanced segmented-vector pulse-frequency modulation (A-SVPFM) strategy and its digital modulator for a half-bridge flying-capacitor three-level converter in wireless power transfer (WPT) systems. The method preserves a zero-switching angle at the resonant frequency across the control range, thereby enabling wide-range zero-voltage switching (ZVS) without auxiliary branches, and was shown to be compatible with practical GaN switching conditions.

In addition, a switching-state selection scheme relying only on voltage sampling was introduced to regulate the flying-capacitor voltage. This approach maintains the capacitor voltage near its reference with small ripple and avoids current sensors, simplifying the sensing and control architecture for multilevel WPT converters.

On the analysis side, the output voltage was formulated via Fourier series and the spectral structure induced by the finite modulation period was clarified: in addition to odd harmonics, the spectrum contains sub-/interharmonic components whose frequencies depend on the modulation period, as captured by Eq. (18). This explains why the interharmonics near the resonant frequency are most influential under the band-pass input impedance of the WPT link. DFT comparisons further showed that the transmitter-current THD of A-SVPFM is markedly lower than that of conventional Δ–Σ PFM.

Moreover, a closed-form ZVS condition was derived that links the required transmitter-capacitance offset, the minimum switching phase/dead time, and the load. The derivation starts from the sinusoidal approximation of the transmitter current during dead time to obtain the minimum phase and dead time, relates the fundamental current to the equivalent input impedance, and finally yields a compact inequality that quantifies the required capacitance margin for ZVS. These results offer practical guidance for device selection and resonant-tank design.

The effectiveness of the proposed scheme was substantiated through analysis and the planned experimental program on a three-level WPT prototype (system topology, ZVS verification, step response, and efficiency/power-loss evaluation). Overall, the contributions—wide-range ZVS via A-SVPFM, voltage-sampling-based capacitor balancing, clarified interharmonics behavior with reduced current THD, and a design-oriented ZVS condition—provide a cohesive pathway toward high-power, sensor-lean multilevel WPT converters.

Looking forward, extensions to higher-level multilevel topologies and multichannel WPT, EMI optimization with interharmonics mitigation, and adaptive scheduling of the resolution factor λ merit further investigation to fully exploit the benefits identified in this work.

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